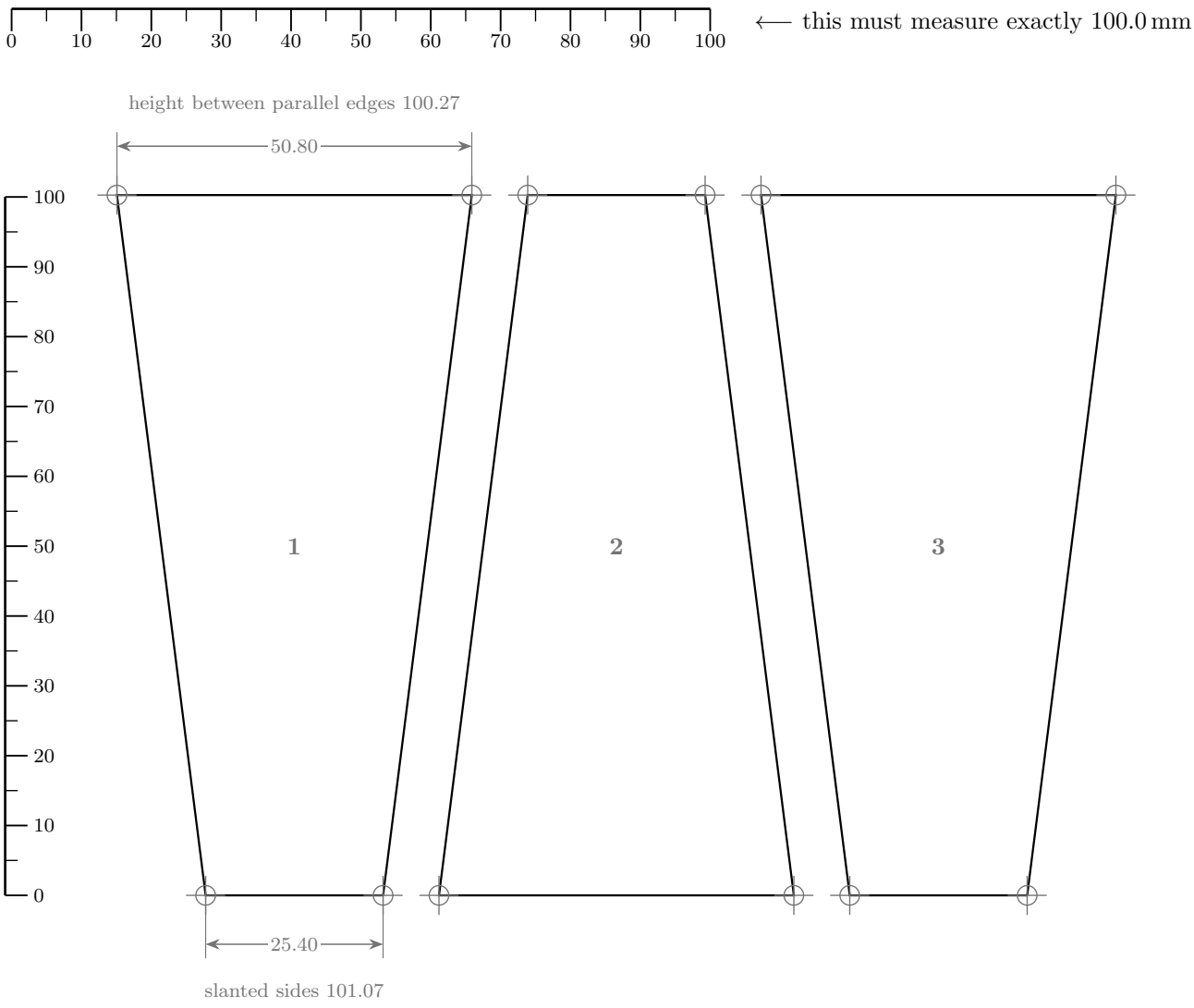
Kaleidoscope mirror cutting template

100 mm prototype · three identical mirrors · drawn 1:1

Print at 100%. In the print dialog choose *Actual size* or *100%* — not *Fit to page*, not *Shrink oversized pages*. Then check *both* rulers below with a real ruler before you cut anything. If they are off, so is every dimension on this page.



How to use it. Leave the protective film on the mirror. Tape this sheet down over it, prick through the centre of all twelve circled marks with a pin or a scribe, lift the paper off, then join the pricks with your straightedge and score along those lines. Marking through the paper is more accurate than trying to cut through it — a knife shreds paper and then wanders.

Cut on the line. These outlines are the finished size and carry no allowance. A scored-and-snapped edge removes essentially nothing, so cut right on the marks. If you ever laser-cut these, offset the toolpath *outward* by half the kerf, or the mirrors come out undersized and leave a gap at every corner.

Only two edges matter. The slanted 101.07 mm sides are the ones that butt against the neighbouring mirrors — they must be straight, and the reflective coating must run right to the edge. The 25.40 and 50.80 mm ends are hidden under the end caps and can be rough.

There is no left or right. Each trapezoid is symmetric, so all three pieces are interchangeable and either face can end up outward. The handedness in this design lives in how the three are *assembled*, not in how they are cut. Mirror 2 is drawn flipped only to nest the three efficiently on the tile.